Quantum-assisted AI Agents Orchestration

Qiskit Fall Fest 2026

Team SpectralTheorem

Meet the team



Kien Nguyen

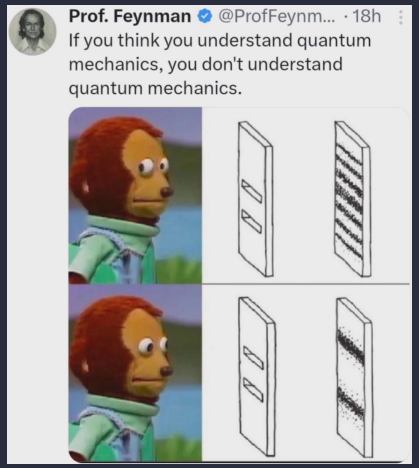
4B Combinatorics and Optimization

Interests:

- Mathematical Modelling
- Operations Research
- Optimization in AI/ ML

Experience in Quantum Computing:

- I watched a few tutorials on Qiskit
- I took a course in Quantum Information
- TL;DR: Basically none :)



What is Multi-agent AI Orchestration?

Multi-agent AI orchestration involves coordinating multiple specialized AI agents to collaboratively accomplish complex tasks. An orchestrator manages task allocation, dependencies, and resource usage to ensure agents work together toward a shared objective.

NP-Hard Complexity

Solving combinatorial task assignments across multiple agents results in NP-hard optimization bottlenecks.

Restrictive Constraints

Managing strict execution deadlines, resource caps, and complex inter-task dependencies.

Dynamic Coordination

Maintaining agent state synchronization and resolving conflicting actions without high latency.

Scalability & Latency

Orchestrating diverse agent capabilities efficiently while preventing system performance decay at scale.

Generalized Assignment Problem (GAP)

A classic **NP-hard** combinatorial optimization model for resource allocation.

Core Objective

Assign a set of N tasks to a set of M agents at minimal cost (or maximum profit).

Each task n is assigned to **exactly one** agent m.

Represented by binary decision variables $x_{nm} \in \{0, 1\}$.

Key Constraints

Capacity Limits: Each agent m has a maximum available budget b_m.

Variable Consumption: Task n consumes a_{nm} resources if processed by agent m.

Heterogeneous agent capabilities create complex trade-offs.

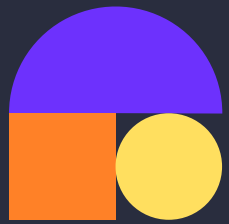
Real-World Applications

AI Orchestration: Multi-agent LLM task routing under token limits.

Cloud Computing: Server load balancing & Virtual Machine placement.

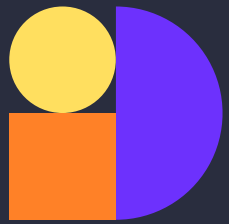
Logistics: Vehicle routing & facility location planning.

Can quantum optimization be applied to resource-constrained task allocation in multi-agent AI systems?



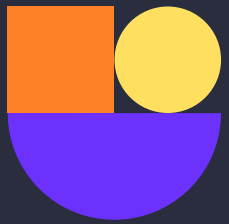
Problem Formulation

Model task allocation across heterogeneous AI agents as a constrained optimization problem.



QUBO Conversion

Map the constrained agent allocation model into a **Quadratic Unconstrained Binary Optimization (QUBO)** formulation with penalty parameters.



VQGAP

Explain intuition behind the **Variational Quantum Generalized Assignment Problem (VQGAP)** algorithm, as well as empirical results

Reference Paper

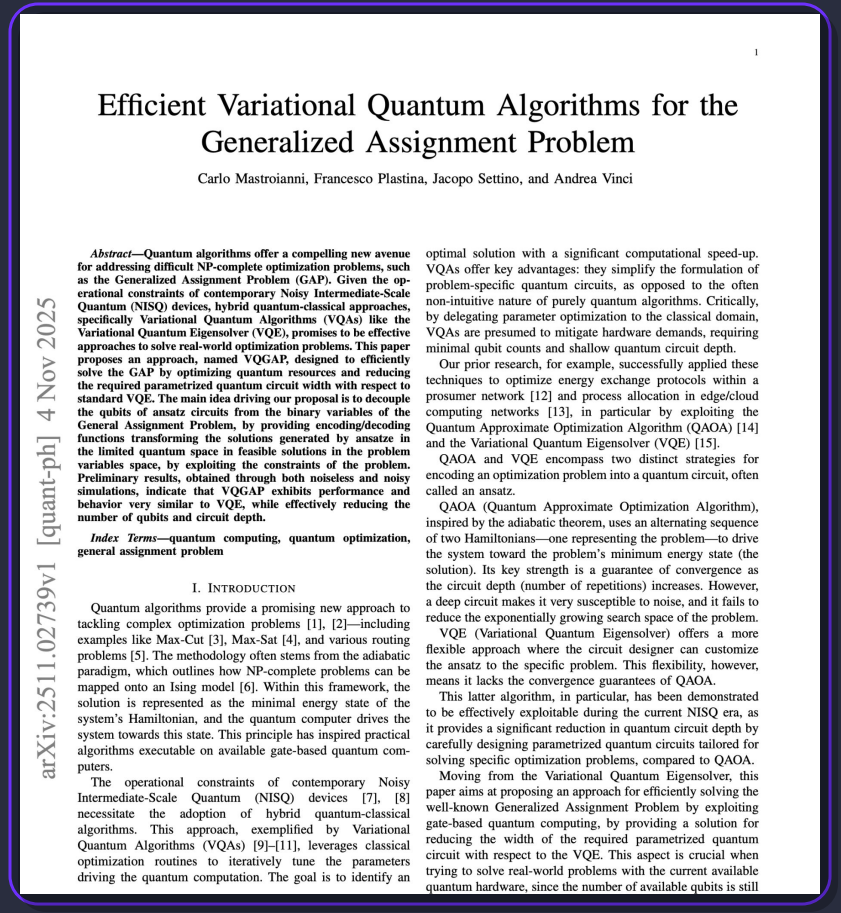
All of the mathematical formulations, algorithms, explanations presented in this project was taken from this phenomenon paper

arXiv:2511.02739

Efficient Variational Quantum Algorithms for the Generalized Assignment Problem

Carlo Mastroianni, Francesco Plastina, Jacopo Settino, and Andrea Vinci

https://arxiv.org/pdf/2511.02739



Efficient Variational Quantum Algorithms for the Generalized Assignment Problem

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arXiv:2511.02739v1 [quant-ph] 4 Nov 2025

***Abstract*—Quantum algorithms offer a compelling new avenue for addressing difficult NP-complete optimization problems, such as the Generalized Assignment Problem (GAP). Given the operational constraints of contemporary Noisy Intermediate-Scale Quantum (NISQ) devices, hybrid quantum-classical approaches, specifically Variational Quantum Algorithms (VQAs) like the Variational Quantum Eigensolver (VQE), promises to be effective approaches to solve real-world optimization problems. This paper proposes an approach, named VQGAP, designed to efficiently solve the GAP by optimizing quantum resources and reducing the required parametrized quantum circuit width with respect to standard VQE. The main idea driving our proposal is to decouple the qubits of ansatz circuits from the binary variables of the General Assignment Problem, by providing encoding/decoding functions transforming the solutions generated by ansatze in the limited quantum space in feasible solutions in the problem variables space, by exploiting the constraints of the problem. Preliminary results, obtained through both noiseless and noisy simulations, indicate that VQGAP exhibits performance and behavior very similar to VQE, while effectively reducing the number of qubits and circuit depth.**

***Index Terms*—quantum computing, quantum optimization, general assignment problem**

I. Introduction

Quantum algorithms provide a promising new approach to tackling complex optimization problems [1], [2]—including examples like Max-Cut [3], Max-Sat [4], and various routing problems [5]. The methodology often stems from the adiabatic paradigm, which outlines how NP-complete problems can be mapped onto an Ising model [6]. Within this framework, the solution is represented as the minimal energy state of the system's Hamiltonian, and the quantum computer drives the system towards this state. This principle has inspired practical algorithms executable on available gate-based quantum computers.

The operational constraints of contemporary Noisy Intermediate-Scale Quantum (NISQ) devices [7], [8] necessitate the adoption of hybrid quantum-classical algorithms. This approach, exemplified by Variational Quantum Algorithms (VQAs) [9]–[11], leverages classical optimization routines to iteratively tune the parameters driving the quantum computation. The goal is to identify an optimal solution with a significant computational speed-up. VQAs offer key advantages: they simplify the formulation of problem-specific quantum circuits, as opposed to the often non-intuitive nature of purely quantum algorithms. Critically, by delegating parameter optimization to the classical domain, VQAs are presumed to mitigate hardware demands, requiring minimal qubit counts and shallow quantum circuit depth.

Our prior research, for example, successfully applied these techniques to optimize energy exchange protocols within a prosumer network [12] and process allocation in edge/cloud computing networks [13], in particular by exploiting the Quantum Approximate Optimization Algorithm (QAOA) [14] and the Variational Quantum Eigensolver (VQE) [15].

QAOA and VQE encompass two distinct strategies for encoding an optimization problem into a quantum circuit, often called an ansatz.

QAOA (Quantum Approximate Optimization Algorithm), inspired by the adiabatic theorem, uses an alternating sequence of two Hamiltonians—one representing the problem—to drive the system toward the problem's minimum energy state (the solution). Its key strength is a guarantee of convergence as the circuit depth (number of repetitions) increases. However, a deep circuit makes it very susceptible to noise, and it fails to reduce the exponentially growing search space of the problem.

VQE (Variational Quantum Eigensolver) offers a more flexible approach where the circuit designer can customize the ansatz to the specific problem. This flexibility, however, means it lacks the convergence guarantees of QAOA.

This latter algorithm, in particular, has been demonstrated to be effectively exploitable during the current NISQ era, as it provides a significant reduction in quantum circuit depth by carefully designing parametrized quantum circuits tailored for solving specific optimization problems, compared to QAOA.

Moving from the Variational Quantum Eigensolver, this paper aims at proposing an approach for efficiently solving the well-known Generalized Assignment Problem by exploiting gate-based quantum computing, by providing a solution for reducing the width of the required parametrized quantum circuit with respect to the VQE. This aspect is crucial when trying to solve real-world problems with the current available quantum hardware, since the number of available qubits is still

Problem Statement

Consider an AI orchestration system with N tasks and M agents.

Each task n is assigned to an agent m, represented by a binary decision variable $x_{nm} \in \{0, 1\}$.

System Parameters & Constraints:

Each agent m has a token budget limit b_m.

Assigning task n to agent m consumes a_{nm} tokens.

Assigning task n to agent m yields an expected success rate $c_{nm} \in [0, 1]$.

To make token costs comparable to success rates, we normalize them by the maximum assignment cost a_{max}.

Goal:

Find an tasks-agent assignment that maximizes the success rate while minimizes the amount of token used

Mathematical Formulation

Our assignment must balance two competing objectives:

1. Maximize Success Rate

$$
\begin{aligned}
\max_x \quad & \sum_{n\in[N],\, m\in[M]} c_{nm}\, x_{nm} \\
\text{s.t.} \quad & \sum_{m\in[M]} x_{nm} = 1, \quad \forall n \in [N] \\
& \sum_{n\in[N]} a_{nm}\, x_{nm} \le b_m, \quad \forall m \in [M] \\
& x_{nm} \in \{0,1\}
\end{aligned}
\tag{P_1}
$$

2. Minimize Tokens Used

$$
\begin{aligned}
\min_x \quad & \sum_{n\in[N],\, m\in[M]} \bar{a}_{nm}\, x_{nm} \\
\text{s.t.} \quad & \sum_{m\in[M]} x_{nm} = 1, \quad \forall n \in [N] \\
& \sum_{n\in[N]} a_{nm}\, x_{nm} \le b_m, \quad \forall m \in [M] \\
& x_{nm} \in \{0,1\}
\end{aligned}
\tag{P_2}
$$

Mathematical Formulation

Since both (P_1) and (P_2) share the same feasible set, we can merge them into a single optimization problem via a weighted sum, with trade-off parameter $\lambda \in [0, 1]$.

Single Optimization Problem

$$
\begin{aligned}
\min_x \quad & \sum_{n\in[N],\, m\in[M]} \left[\lambda\, \bar{a}_{nm} - (1-\lambda)\, c_{nm}\right] x_{nm} \\
\text{s.t.} \quad & \sum_{m\in[M]} x_{nm} = 1, \quad \forall n \in [N] \qquad \text{(each task assigned once)} \\
& \sum_{n\in[N]} a_{nm}\, x_{nm} \le b_m, \quad \forall m \in [M] \qquad \text{(token budget)} \\
& x_{nm} \in \{0,1\}, \quad \forall n \in [N], \forall m \in [M]
\end{aligned}
\tag{P_{GAP}}
$$

QUBO Formulation

Since VQE and VQGAP only accepts combinatorial optimization problems in a quadratic unconstrained binary optimization problem, we must convert **(PGAP)** to **(PQUBO)**.

QUBO Formulation

Defining the combined weight

$$w_{nm} = \lambda \bar{a}_{nm} - (1-\lambda) c_{nm}, \quad \forall n \in [N], \forall m \in [M]$$

the standard approach converts the constraints into quadratic penalty terms:

$$H(x,y) = \sum_{n\in[N],\ m\in[M]} w_{nm} x_{nm} + C \sum_{n\in[N]} (1 - \textstyle\sum_m x_{nm})^2 + C \sum_{m\in[M]} (b_m - \textstyle\sum_n a_{nm} x_{nm} - s_m)^2$$

where

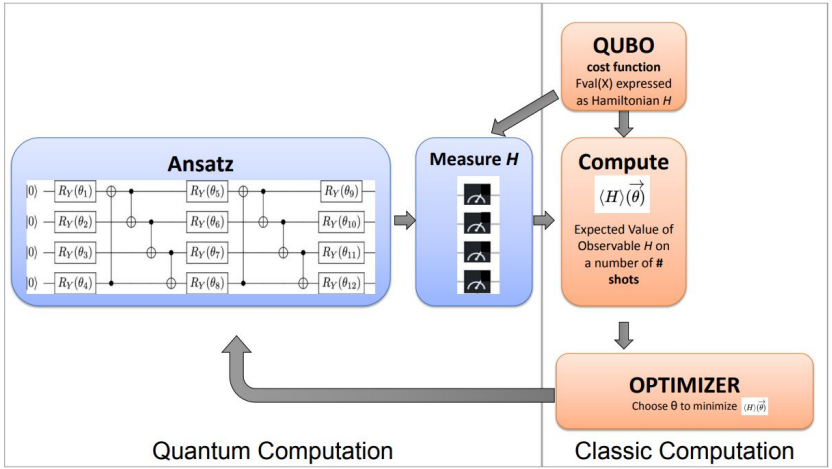
$$C = 1 + \sum_{n,m} |w_{nm}|$$

Further, in the above formulation, each slack variable $s_m, \forall m \in [M]$ represents the unused token budget of agent m.

Variational Quantum Eigensolver (VQE)

Solves optimization problems by minimizing the Hamiltonian **H(x,y)**, but requires expressing all problem variables as **qubits**.

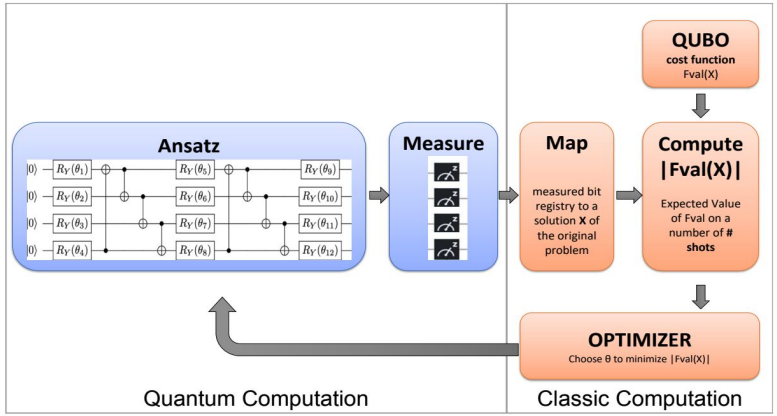
Hybrid VQE Algorithm Loop



Variational Quantum Generalized Assignment Problem (VQGAP)

Similar to VQR, but offloads the residual budget calculations to a classical computation ⇒ Uses less qubits

Hybrid VQGAP Algorithm Loop



Jupyter Notebook

Any questions? Ask away!

Thank you